

What are all the information I need to upload into the Kairo Design Project as I start the planning phase for the product design. Note: I already had a product called Robomed which from our conversation you know how and what it's been serving students with. Students have had their praises and critics and I'm trying to revamp the experience by building a new and better infrastructure independently and integrating the students' data from supabase and giving them an even better experience from the moment they visit the site. Remember: the bigger picture isn't for them to be able to practice question but a dynamic user-centric learning experience that solves the real problems of Nigerian students preparing for UTME with the exams not just the only focus but the becoming. Retention is the one of the most critical factors so we make this a progressive journey where their memory is reinforced daily using spaced repetition and a recap using personalised sessions daily before they can move on with attempting other sessions with different options available to them like the ones we designed for Robomed such as Rapid Fire, Custom Subject Practice, Topic-by-topic practice, contextual AI assistant embedded in the platform, learning journey explicitly visible in their profiles' homepage, etc.

Note: we're not just building it with profit at heart but using the most sustainable ways to build a high student-centered platform that grows with their experiences, suggestions and pain points to become their best companion always available to them as they progress.

It'll definitely have a dynamic, segmented leaderboard of 20 people each which will be based on their category of students with similar strengths and weaknesses rather than a leaderboard comprising of everyone where they feel oppressed. There'll be university leaderboard too so they can see and help their aspiring university grow on the leaderboard through their practice sessions. There'll be levels, badges, etc. I am still learning using the first product to better understand students and what they need from us.

One of the greatest problems the students had was the onboarding, they didn't even understand what to do and how to go about it from the moment they visited the platform. We have to keep it simple, sleek and very endearing to them. I'm obsessed with the quality of their experience and want this product to be built on that since it's a habit-forming platform designed to make them love education.

Under their learning journey tab, dedicated and personalised practice sessions will be available under the different specific weaknesses observed in the course of their learning to foster retention and understanding.

Good evening. Yes—absolutely. What you are describing is not only possible; for Kairo, it may actually be one of the most strategically important architectural decisions you can make.

The key idea is:

> Kairo does not need to be “an online website that sometimes works offline.” It can be designed as an offline-first web application.

That is a very different mindset.

The model I would recommend for Kairo

Think of Kairo as having two layers:

1. The Core Learning Engine — works offline

Once a student has downloaded the necessary content, they should be able to do things like:

Answer UTME questions

Take mock tests

Review explanations

Read downloaded lesson materials

Track progress

See previously downloaded performance data

Use certain revision tools

Continue answering questions even when there is no network

The student could literally:

> Open Kairo → lose internet connection → continue studying.

This is possible using a Progressive Web App (PWA) architecture.

2. The Synchronization Layer — requires internet occasionally

When internet is available, Kairo can:

Download new questions

Download new lessons

Synchronize answers

Upload test results

Update the leaderboard

Sync streaks and achievements

Download new features or app updates

Update the student's account across devices

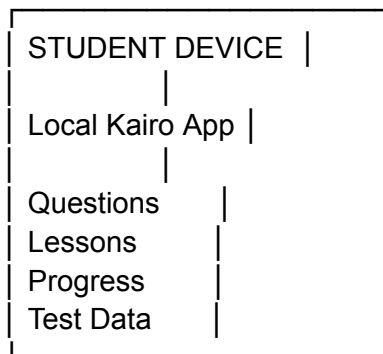
So instead of requiring internet for every action, Kairo only requires internet when data needs to move between the student's device and the server.

The basic flow becomes:

INTERNET AVAILABLE



Download / Synchronize



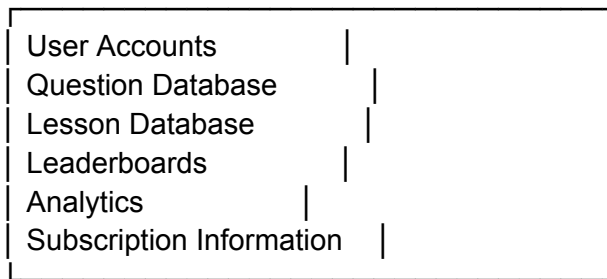
INTERNET LOST

↓
Student continues
studying offline

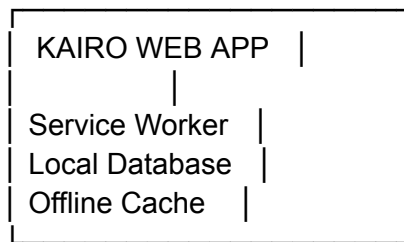
The important thing is that the device becomes a temporary local copy of the student's Kairo environment.

The architecture would look something like this

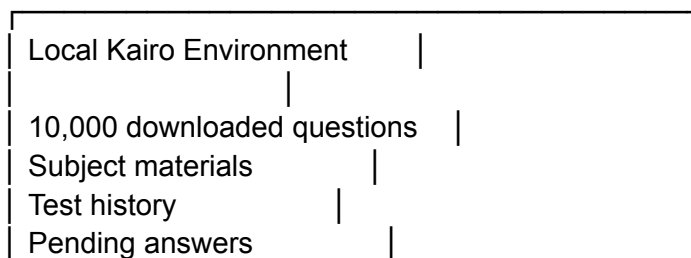
KAIRO CLOUD SERVER

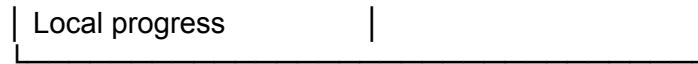


INTERNET



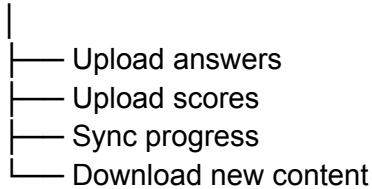
STUDENT DEVICE





The student may use Kairo for several days without internet. Then, when they get access to Wi-Fi or affordable data:

LOCAL DATA



But here is the important distinction

There are different levels of "offline."

Level 1: Offline website caching

You cache the basic interface.

The student can open Kairo without internet, but only limited things work.

This is the simplest version.

For example:

Homepage loads

Some pages load

Previously visited content opens

But this is not enough for your vision.

Level 2: Offline learning

This is much more useful.

The student deliberately downloads:

> "Biology: Cell Biology — 500 questions"

or:

> "Physics: Mechanics — Complete Revision Pack"

Then they can study that material without internet.

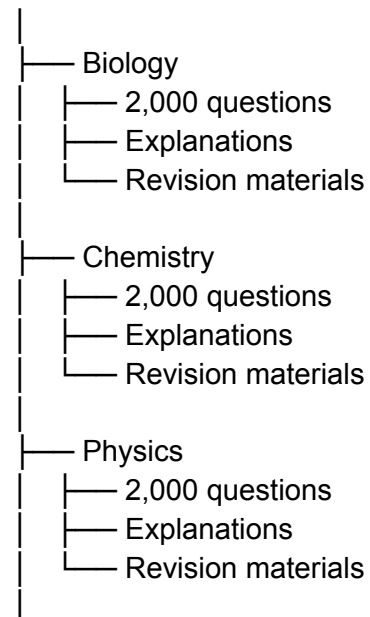
This is probably where Kairo should begin.

Level 3: Offline-first Kairo

This is the more ambitious version.

The student downloads a substantial portion of their learning environment:

Kairo



- Local progress
- Test history
- Streak
- Pending synchronization

The student can then use Kairo as a study application, not merely as a website.

This is the direction I think you should seriously consider.

The technology behind it

You can build this using web technologies.

Frontend

For example:

React

Next.js

Vue

Svelte

Offline capability

Using:

Service Workers

Cache Storage

IndexedDB

Local-first data architecture

Backend

Your existing backend could handle:

Authentication

Questions

User accounts

Progress

Analytics

Synchronization

The important technology is the local database.

A student might answer 100 questions while offline.

Instead of immediately sending every answer to your server:

Question 1 → Internet → Server

Question 2 → Internet → Server

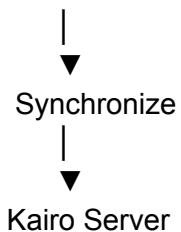
Question 3 → Internet → Server

You do:



Then later:

Internet becomes available



This is particularly useful for Nigerian students because data availability is not always consistent.

A student might have ₦100 worth of data today, no data tomorrow, and access to Wi-Fi next week.

Kairo should not punish them for temporarily being offline.

But I would not make the entire app offline from Day One

This is where I would challenge your thinking slightly.

You do not need to build the most complex possible offline architecture immediately.

You are still validating Kairo.

I would build it in phases.

Phase 1: The minimum viable offline experience

Allow students to:

1. Download question packs
2. Download explanations
3. Download revision materials
4. Take tests offline
5. Save results locally
6. Synchronize when internet returns

That alone could be a powerful differentiator.

Phase 2: Intelligent synchronization

Later, Kairo can intelligently synchronize:

Student completes 50 questions offline



Results saved locally



Internet returns



Kairo uploads results



Progress updates



Leaderboard updates



Achievements unlock

You could even show:

> Last synchronized: 2 days ago

and:

> 17 activities waiting to synchronize

This creates transparency and prevents students from wondering whether their progress was lost.

One major challenge: data storage

You need to think carefully about how much content you want to store locally.

Suppose:

1 question = 2 KB

10,000 questions = approximately 20 MB

That is relatively manageable.

But if every question contains:

Images

Videos

Audio

Large explanations

the storage requirements increase significantly.

Therefore, I would recommend:

Store locally:

Questions

Answers

Explanations

Student progress

Test history

Stream when possible:

Videos

Large audio files

Heavy images

Or allow them to be separately downloaded.

The most important architectural principle

I think Kairo should be designed around this principle:

> The internet should enhance the experience, not determine whether the student can study.

That is a powerful product philosophy.

For a Nigerian UTME candidate, the reality might be:

Monday:

Internet available

Tuesday:

No data

Wednesday:

Poor network

Thursday:

Uses school Wi-Fi

Friday:

No internet again

A normal web platform says:

> "You cannot study."

An offline-first Kairo says:

> "Your learning continues. We'll synchronize when you reconnect."

That is a fundamentally better product for your target market.

My recommendation for your research phase

Before building this, I would investigate four things with actual Nigerian UTME students:

1. How frequently do they lose internet access?

Ask:

> "How often do you study without internet access?"

2. What do they currently do when they have no data?

Do they:

Read PDFs?

Use downloaded apps?

Use WhatsApp materials?

Use screenshots?

Stop studying?

3. What would they download?

Would they prefer:

> "Download Biology"

or:

> "Download 500 questions"

or:

> "Download today's study plan"

This matters.

4. How much storage do their phones have?

This is very important.

An offline-first architecture is only useful if students have enough storage to use it.

My honest opinion

For Kairo, I would not rush into building a native Android app simply because offline capability is important.

A well-designed Progressive Web App can give you:

A web-based product

Installability on Android

Offline capability

Lower development overhead

Easier updates

One codebase

Better control during validation

The key is to build Kairo as an offline-capable application from the beginning, rather than building a normal online web app and trying to bolt offline functionality onto it later.

That architectural decision will affect your data model, synchronization logic, question delivery system, and frontend structure.

So yes: you can absolutely build Kairo as a web app that functions substantially offline. And given the realities of Nigerian students, I think the right question is not "Can we make Kairo offline?"

The better question is:

> Which parts of Kairo must work offline, which parts should work offline, and which parts genuinely require the internet?

That distinction will help you build something intentional rather than just adding an "offline mode" as a feature.